MID SEMESTER DESIGN REVIEW

PaveWatch

UAV inspection of paved-surface geometry and visible distress

Aerovengers · Mississippi State University

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Faculty advisor: Jingdao Chen

Team and responsibilities



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Diagnostics & Telemetry



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Sensor Payload & Data Acquisition



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Flight Platform, Power & Control



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Ground Control & Communications



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Data Processing & Change Detection



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Reporting, Database & Alerts

Faculty advisor: Jingdao Chen

Pavement inspection problem

1,700+

sidewalk injury claims

>$35 million

in settlements

Los Angeles · five fiscal years [1]

Intended users

Campus facilities and municipal maintenance teams

Inspection need

Repeatable locations, images, and surface measurements for follow-up inspection and maintenance records.

References: [1], [2]

Inspection approaches

Approach	Primary information	PaveWatch design contribution
Manual inspection	Direct observation and inspector judgment	Location-linked records for follow-up inspection
RGB imaging	Visible distress, texture, and context	Pair images with laser-derived surface measurements
Photogrammetry	Image-derived surface reconstruction	Use direct range profiles as a geometry input
PaveWatch	Laser profiles + images + aircraft pose	A referenced survey package and maintenance review interface

References: [2]

Design features

F1 · Supervised survey

Define a paved area and flight route; supervise collection from the ground station.

F2 · Surface assessment

Identify broad raised or depressed areas, steps, gradients, and visible distress.

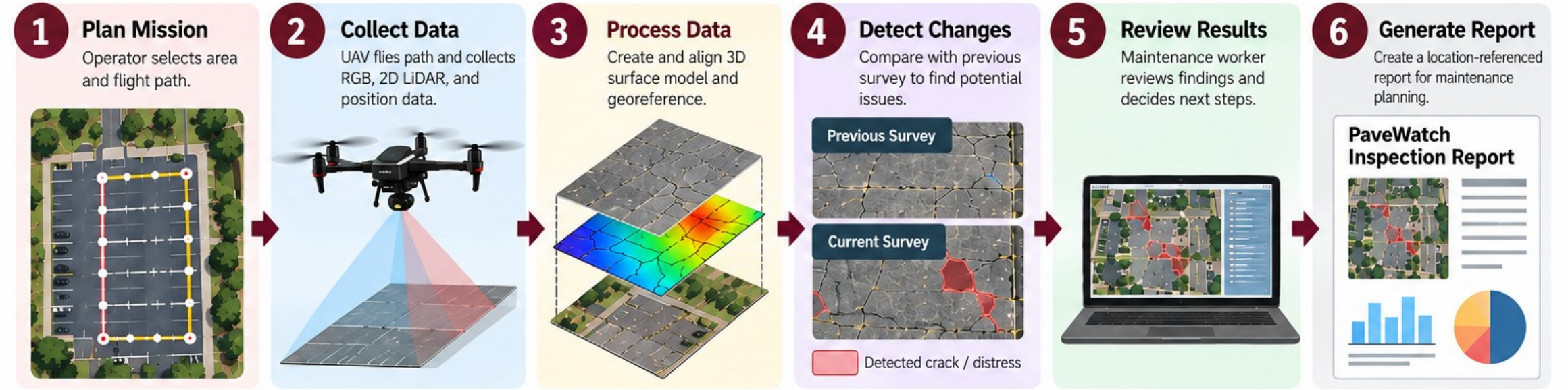
F3 · Evidence review

Inspect images, available measurements, units, and data-quality warnings for each finding.

F4 · Referenced data

Combine position, orientation, timestamps, and sensor mounts in a common survey frame.

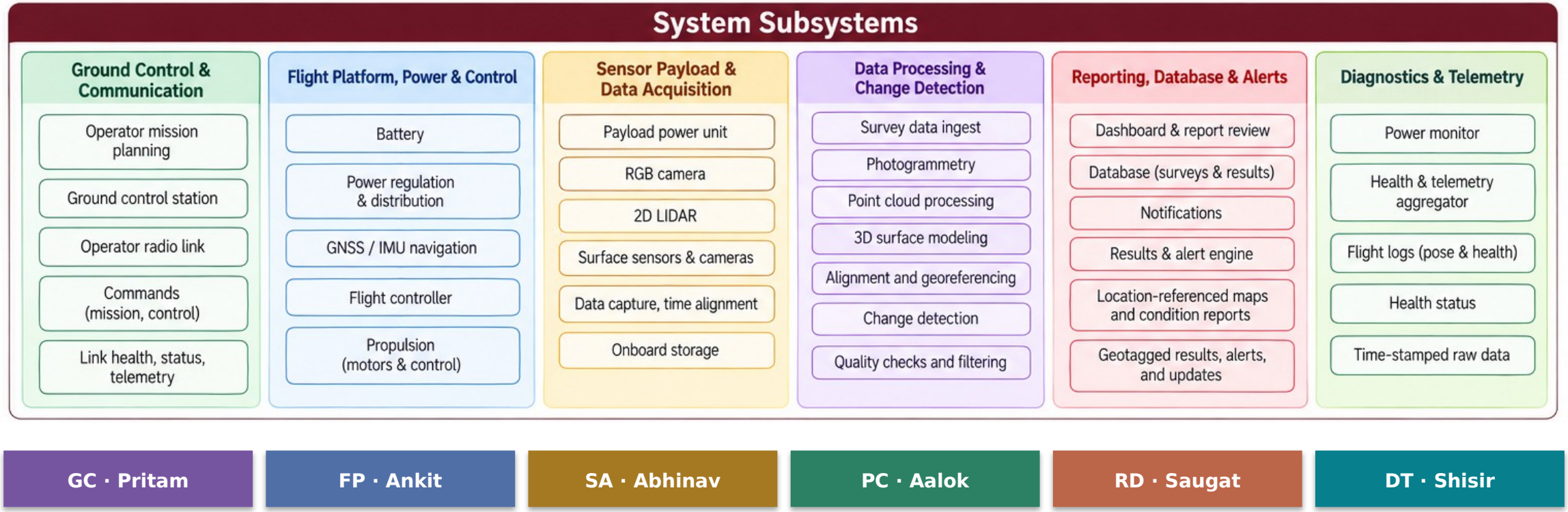
Survey workflow



Current-survey geometry and visible distress form the baseline.

Previous-survey comparison is an optional extension.

Subsystem architecture



Onboard: laser profiles, images, pose, and timestamps.

After landing: transfer → processing → evidence review and report.

System requirements: flight and geometry

Feature / ID	Acceptance target	Evaluation
F1 / S1	≥10 min flight with acquisition active; ≥20% calibrated usable energy remaining at landing	Three loaded flights
F2 / S2	100–150 mm geometry in regions ≥0.5 × 0.5 m; recall ≥90%, precision ≥80%; report gradients	≥10 feature cases and ≥10 negative regions
F2 / S3	≥90% of detected targets jointly meet height error ≤60 mm and centroid error ≤0.5 m	Independent height and location references; height uncertainty ≤10 mm

System requirements: data and imagery

Feature / ID	Acceptance target	Evaluation
F3 / S4	Record range and images for ≥10 min; identify stream gaps >1 s	Concurrent recording and controlled gap injection
F2–F3 / S5	Per-class precision and recall ≥80%; crack-like targets ≥20 mm × 0.2 m; depressions ≥0.5 × 0.5 m	≥10 positives per class and ≥10 nuisance-only regions
F4 / S6	Record 3D position and orientation for ≥10 min; reference each quantitative observation	Clock/frame/mount checks and missing-reference tests

Operating and integration constraints

Area	Planning bound	Design response
Survey operation	5–8 m above pavement; 1–2 m/s; dry daylight	Plan route spacing, exposure, and scan coverage together
Flight/site	Part 107 operating case; VLOS; <55 lb; ≤400 ft AGL; campus request ≥3 weeks ahead	Schedule site authorization before field trials
Aircraft budget	Under $700 for aircraft, flight battery, RC, and navigation; recording payload excluded	Compare complete aircraft configurations and delivered quotes
Payload installation	650 g target; 800 g upper planning case; picoScan supply 9–30 V DC	Allocate mass, mounting space, and separate payload power branches

References: [13], [14], [15]

Applicable standards

Standard	Application	Verification
IEEE 802.3-2022	Ethernet connection from LiDAR to recorder	Verify link configuration and ≥10 min capture; inspect packet gaps
IETF RFC 7946	Geographic finding exports in GeoJSON	Parse ≥10 findings; verify WGS 84 longitude/latitude ordering
IETF RFC 8446	TLS 1.3 for network upload and report access	Verify negotiated version and certificate handling

References: [16], [17], [18]

Flight-system options

Configuration	Planned cost	Selection attributes
F450 DIY + SpeedyBee F405 V4	$405 estimate	Custom assembly; F9P rover navigation concept; FlySky RC; 3S/4S battery selection
X500 v2 ARF + Kakute H7 Mini v1.5	$657.17 subtotal	250 mm payload rails; matched propulsion; M10 GNSS/compass; SiK telemetry; 4S 5 Ah
S500 v2 kit + Pixhawk 6C	$669.20 subtotal	Integrated flight electronics; M10 navigation; SiK telemetry; matched propulsion and 4S 5 Ah

Planning choice: X500 / Kakute for its documented payload-mounting interface.

References: [3], [4], [5], [6], [24]

Recorder options

Configuration	Interfaces / allowance	Selection attributes
Orange Pi Zero 3 + expansion board	Gigabit Ethernet; USB 2.0; approximately $50 for board + expansion	Compact recorder for LiDAR, USB camera, and external SSD
Raspberry Pi 4 · 2 GB	Gigabit Ethernet; USB 3.0; CSI; approximately $60 board allowance	Native USB ports and storage interface; CSI camera option
Pi Zero 2 W + Ethernet/USB hub	CSI; USB host with Ethernet adapter; approximately $35 board + hub	Compact modular recorder with shared USB connection

Planning choice: Orange Pi Zero 3 with USB camera and external SSD.

References: [7], [8], [9]

Camera options

Camera	Image / interface	Selection attributes
InnoMaker U20CAM-1080P	Color; USB 2.0 UVC; $18.69 listed	Budget color recording; verify negotiated 1080p mode and lens coverage
InnoMaker U20CAM-9281M	1280 × 800 monochrome; global shutter; $32.99 listed	External trigger; manual exposure; grayscale analysis path
Arducam B0495 / AR0234	1920 × 1200 color; global shutter; USB 3.0	Color with global exposure; pair with a USB 3 recorder for full mode

Planning choice: InnoMaker color USB camera; select lens and exposure for the survey envelope.

References: [10], [11], [12]

Payload installation and power

650 g

Survey payload target

800 g

Upper planning case

4S / 5 Ah

Flight battery plan

Mechanical layout

Sensor module: 120 × 100 × 130 mm

Recorder module: 120 × 90 × 60 mm

Rigid LiDAR/camera mount; measured center of gravity

Power distribution

4S battery → propulsion / flight power

Fused sensor branch → picoScan

Dedicated 5 V regulator → recorder, camera, SSD

Packaging dimensions and masses are design allocations.

References: [3], [13], [23]

Aircraft budget

X500 / Kakute aircraft	Cost
X500 ARF frame and matched propulsion	$329.00
Kakute H7 Mini + M10 + SiK pair + PM02	$187.97
RC + 4S battery + adapter + transmitter cells	$110.20
Wiring and assembly consumables	$30.00

$657.17

Aircraft parts subtotal

$42.83

Allowance below $700

Payload budget: recorder + expansion + camera ≈ $68.69; add storage, supplies, cables, and mount.

Aircraft subtotal excludes shipping/tax and a new charger. Recording payload is a separate cost scope.

References: [3], [6], [19], [20], [21], [22], [23]

Project development

Competition participation

Selected for the SICK$10K Challenge with a competition-provided picoScan sensor.

System definition

Six subsystem owners; quantitative flight, geometry, recording, imagery, and pose requirements.

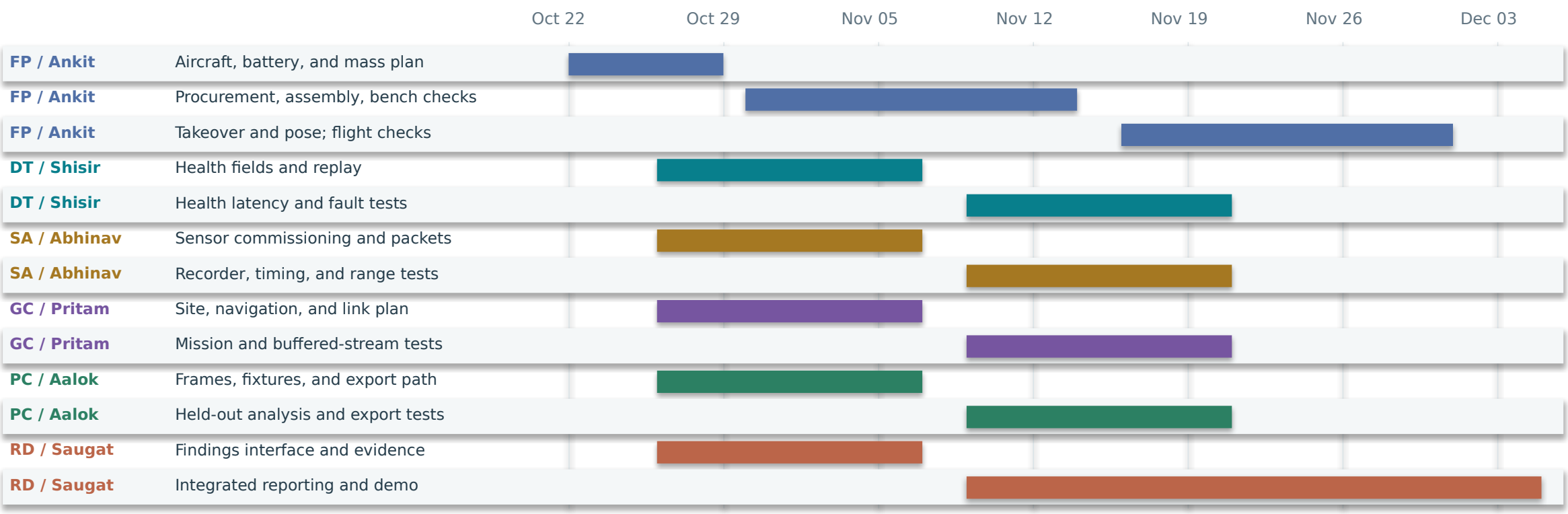
Hardware planning

Three flight-system choices; recorder and camera comparisons; payload mass and power allocations.

Next integration milestones

Commission the sensor, implement simultaneous recording, connect health reporting, and evaluate reference targets.

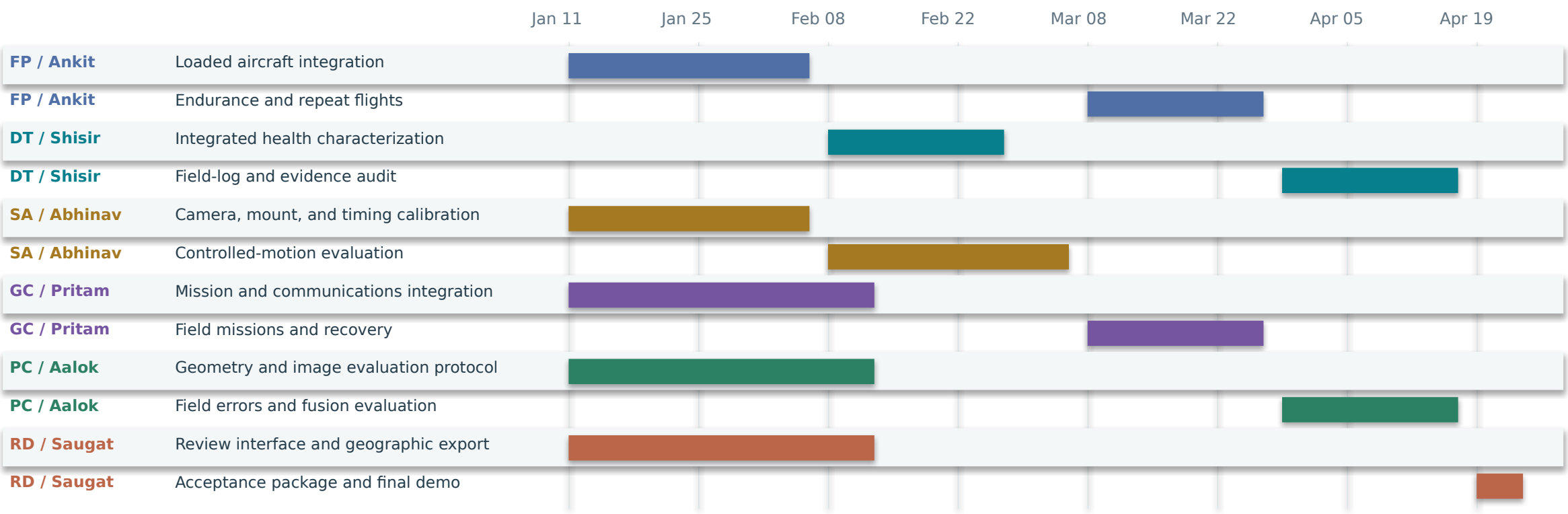
Capstone I schedule



Oct 30 · Approach draft **Nov 13 · Approach final** **Nov 20 · Demo rubric**

Nov 30 · Demonstration **Dec 4 · Final document**

Capstone II schedule



Proposed dates · Jan 11–Apr 23, 2027

Integration → calibration → field evaluation → acceptance and release

Capstone I demonstration deliverables

Subsystem / owner	Performance criteria	Method and delivered artifact
Sensor Acquisition **Abhinav Dhakal**	≥10 min range + images; gaps >1 s flagged; residual timing ≤20 ms; ≥90% range errors ≤60 mm	Bench recording at 5, 6.5, 8 m; ≥10 observations/condition; independent timing events. Data package + plots.
Diagnostics & Telemetry **Shisir Baral**	≥90% health updates ≤4 s for ≥10 min; sensor/pose loss warnings	≥300 updates; inject 5 gaps per stream; disconnect pose/sensor. Status view + latency/fault logs.
Data Processing **Aalok Uprety**	≥90% frame checks: ≤1 m X/Y, ≤0.2 m Z; S2/S3/S5 metrics; 3 exports ≥1 GB, each ≤30 min	Three known poses + ≥10 check points; held-out geometry/image cases; integrity checks. Analysis + findings + exports.

Demonstration package: recorded survey, health evidence, processing results, and repeatable test procedures.

Verification plan

Surface geometry

Use known 100–150 mm features, regions ≥0.5 × 0.5 m, and independent height/location references.

Timing and pose

Check events across the recording; verify common frames, mount transforms, and ≤5° attitude error in ≥90% of samples.

Detection performance

Reserve evaluation cases before tuning. Report counts, precision, recall, joint S3 passes, and failure cases.

Integrated operation

Measure three loaded flights, route tracking, operator takeover, power reserve, and end-to-end evidence integrity.

Integration decisions

Decision	Planned definition	Lead
Mechanical and power	Freeze installed mass, sensor clearance, mount geometry, 4S charging, and the final delivered quote	Ankit + Abhinav
Navigation and timing	Document GNSS/IMU pose, clocks, mounting transforms, and independent reference points	Ankit + Aalok
Ground communication	Define GC3's 60 s buffered stream, data rate, reconnect ordering, and operator health fields	Pritam + Shisir
Evaluation and reporting	Agree gradient-error criterion, evidence fields, geographic export, and reviewer warnings	Aalok + Saugat

Project Manager: overall review, architecture, budget, scope, and final release decisions.

Future work and broader impact

Capstone I

Commission acquisition, demonstrate data-quality monitoring, and produce a traceable processing package.

Capstone II

Integrate the loaded aircraft and evaluate surface accuracy, imagery, and repeatability in field trials.

Maintenance use

Provide campus and public-works staff with location-linked evidence for targeted follow-up and condition records.

Further development

Evaluate repeat-survey changes, paired image/range analysis, and the time and cost of the inspection workflow.

References: inspection and aircraft

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